Higher Colleges of Technology

كليات التقنية العليا

Final Assessment-Alternative Assessment

Semester	202520	**Division**	CIS
Assessment title in Syllabus	**Final_Project_Portfolio**	**Program**	Bachelor of Information Technology/ Bachelor of Information Systems

Course Code	CDS 2413		
Course Title	INTRODUCTION TO DATA SCIENCE		
Course Instructor		**CRN**	
Assessment	Project Deliverable 1 – Data Analysis Report	**Submission Date**	**Week 6**

Student Declaration:

Academic Integrity Statement

In accordance with the HCT Academic Integrity Policy

- Students are required to refrain from all forms of academic integrity breaches as defined and explained by HCT.
- A student found guilty of having committed acts of academic integrity breach(es) will be subject to the relevant sanctions as outlined by HCT.

إفادة النزاهة الأكاديمية

وفقًا لسياسة كليات التقنية العليا للنزاهة الأكاديمية

- على الطلبة الإلتزام بلوائح وقواعد النزاهة الأكاديمية، كما هو مبيّن وموضح في السياسات والإجراءات الخاصة بكليات التقنية العليا.
- في حالة ارتكاب الطالب أي شكل من أشكال الإخلال بالنزاهة الأكاديمية، سيتعرض الى العقوبات الموضحة في السياسات ذات الصلة.

This assignment is entirely my/our own work except where I have duly acknowledged other sources in the text and listed those sources at the end of the assignment. I have not previously submitted this work to the HCT, or any other entity. I understand that I may be orally examined on my submission.

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Student Signature				

Chatbot (e.g. ChatGPT) statement

This submission allows you to use AI generated tools, but the prompt used must be stated and the response must be in red font. Include References to ChatGPT.

Instructions

Remove instructions and hints from the template before submitting the final report to BBLearn.

CLO1: Introduction and Data Wrangling (5 Points)

Video games sales with rating
https://www.kaggle.com/datasets/rush4ratio/video-game-sales-with-ratings

Purpose of Data Analysis (1 point)

<< Define the purpose of data analysis for the chosen dataset >>.

The purpose of analyzing this video game sales dataset is to understand what affects games global sales. By looking at things like scores, user ratings, and platform, we can find patterns and learn why some games sell more than others.

Type of Programming (1 point)

<< Identify and Justify the type of programming used for data analysis.>>

We are using Python for this project, such as Pandas, Matplotlib, and Seaborn.

Type and Purpose of ML algorithm (1 point)

<< Identify the type and purpose of the machine learning algorithm to be implemented for the chosen dataset >>

We will use regression for values like global sales and classification to categorize games like high vs low sales.

Identify the Varaibles (1 point)

<< Identify and Justify the independent and dependent variables for the chosen dataset.>>

The dependent variable is Global_Sales because it is the value we want to predict
The Independent variable include Critic_Score, User_Score , Genre , Rating , and Year_of_Release.

Sampling (1 point)

<< Will you do the sampling? Identify and Justify the type of sampling to be used for the chosen dataset.>>

Yes, we will use sampling
Random sampling ensures that every data point has an equal chance of being selected, and Systematic sampling helps in selecting data at regular intervals and makes it simple and efficient for large datasets.

CLO2: Descriptive, Exploratory and Statistical Analysis (10 Points)

Purpose of Descriptive Analysis (1 point)

<< Justify why you want to perform the descriptive analysis for the chosen dataset. >>.

It helps to understand the basic behavior of Global_sales, and it shows the average sales, and it gives clear ideas of the dataset.

Script for Descriptive Analysis (1 point)

<< Create a script to develop a Python function for descriptive statistics. The input for the function should be the sample and the field to perform the descriptive statistics..>>

```
def descriptive_stats(data, column):
    stats = {
        "Mean": data[column].mean(),
        "Median": data[column].median(),
        "Mode": data[column].mode()[0],
        "Std": data[column].std(),
        "Min": data[column].min(),
        "Max": data[column].max()
    }
    return stats
```

Random Sampling(1 point)

<< Create a program to random sampling of size 150 and find the descriptive statistics for the dependent variable from the sample [Apply the descriptive function which you created].>>

```
sample_random = df.sample(150)
descriptive_stats(sample_random, "Global_Sales")
```

Systematic Sampling (1 point)

<< Create a script for systematic sampling by giving certain conditions and finding the desc stat for the dependent variable from the sample [Apply the descriptive function which you created]..>>

```
sample_systematic = df.iloc[::10]
descriptive_stats(sample_systematic, "Global_Sales")
```

Descriptive Statistical Report (1 point)

<< Create a detailed descriptive statistics report about the dependent variable of the chosen dataset.>>

The Global_Sales is mostly low for most games, only a few games have very high sales. This makes the data skewed because a small number of games are big hits.

Visualisation through Various Charts (2 points)

1. << Visualize the dependent variable by the Graph/Chart of the following using Python Program:
    a. Scatter plot
    b. Box Plot
    c. Histogram
    d. Heat Map

Hint: Use Matplot or Ski-learn library.>>

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.scatterplot(data=df, x="Critic_Score", y="Global_Sales")
plt.title("Critic Score vs Global Sales")
plt.show()
```

```
sns.boxplot(data=df, y="Global_Sales")
plt.title("Box Plot of Global Sales")
plt.show()

plt.hist(df["Global_Sales"], bins=30)
plt.title("Global Sales Distribution")
plt.show()

sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

Hypothesis Test (1 point)

<< 8. Perform the hypothesis test to find the correlation (Pearson and Spearman for numerical variable and chi-square test for categorical variable) between the independent variable and the dependent variable.
Note: If you have more than one independent variable, then choose any one of the independent variables..>>

```
from scipy.stats import pearsonr, spearmanr

corr_p, p_value_p = pearsonr(df["Critic_Score"], df["Global_Sales"])
corr_s, p_value_s = spearmanr(df["Critic_Score"], df["Global_Sales"])
```

if p-value is less than 0.05, there is a relationship between critic score and global sales
if p-value is greater than 0.05, there is no relationship

One Sample T-test (2 points)

<< Assess the performance of the dependent variable to know whether the sample is representative of the normal population by a one-sample t-test..>>

```
from scipy.stats import ttest_1samp

sample = df.sample(150)

t_stat, p_value = ttest_1samp(sample["Global_Sales"], df["Global_Sales"].mean())
```

if p-value is less than 0.05, the sample mean is different from the population mean
if p-value is greater than 0.05, the sample represents the population well

This helps us to understand whether critic score affect global sales

End of Documentation.